

1. Introduction

The Bank Account Management System is a Java application that uses a command line interface to model how an automated teller machine functions. The primary goal of this software is to execute standard financial tasks, including the creation of accounts, the addition of funds, the removal of funds and the display of account information. I constructed this project to apply fundamental object oriented programming methods and to gain experience with Java Collections.

2. Objectives

The project aims to create an interactive application for the console. It exists so that I can define plus use classes, objects and methods. I intend to use a Hash Map to store and access user information within the computer memory. To manage how the software reacts to user entries, I used loops but also switch statements.

3. Technologies Used

Programming Language: Java (JDK 8 or higher)

Core Concepts: Object-Oriented Programming, Java Collections Framework (HashMap)

Input/Output: `java.util.Scanner`

Environment: Command Line Terminal / IDE (VS Code, IntelliJ, etc.)

4. Project Modules

The project is divided into the following key operations, managed through a continuous do-while loop:

- i. Open Account: Takes the customer's name and an initial deposit amount. It generates a unique account number starting from 1001 and stores the Account object in a HashMap.
- ii. Deposit Money: Asks for the account number, validates if it exists, and adds the specified amount to the account balance.
- iii. Withdraw Money: Verifies the account number and checks if the balance is sufficient before allowing the withdrawal. It prevents negative balance scenarios.
- iv. Check Account Details: Retrieves the account using the account number and displays the customer name, account number, and current balance.
- v. Exit: Safely breaks the loop and closes the application.

5. Output

```
1. Open Account
2. Deposit Money
3. Withdraw Money
4. Check Account Details
5. Exit
Enter your choice: 2
Enter account number: 1001
Enter amount to deposit: 1500
Deposit successful!
New balance: 3800.0

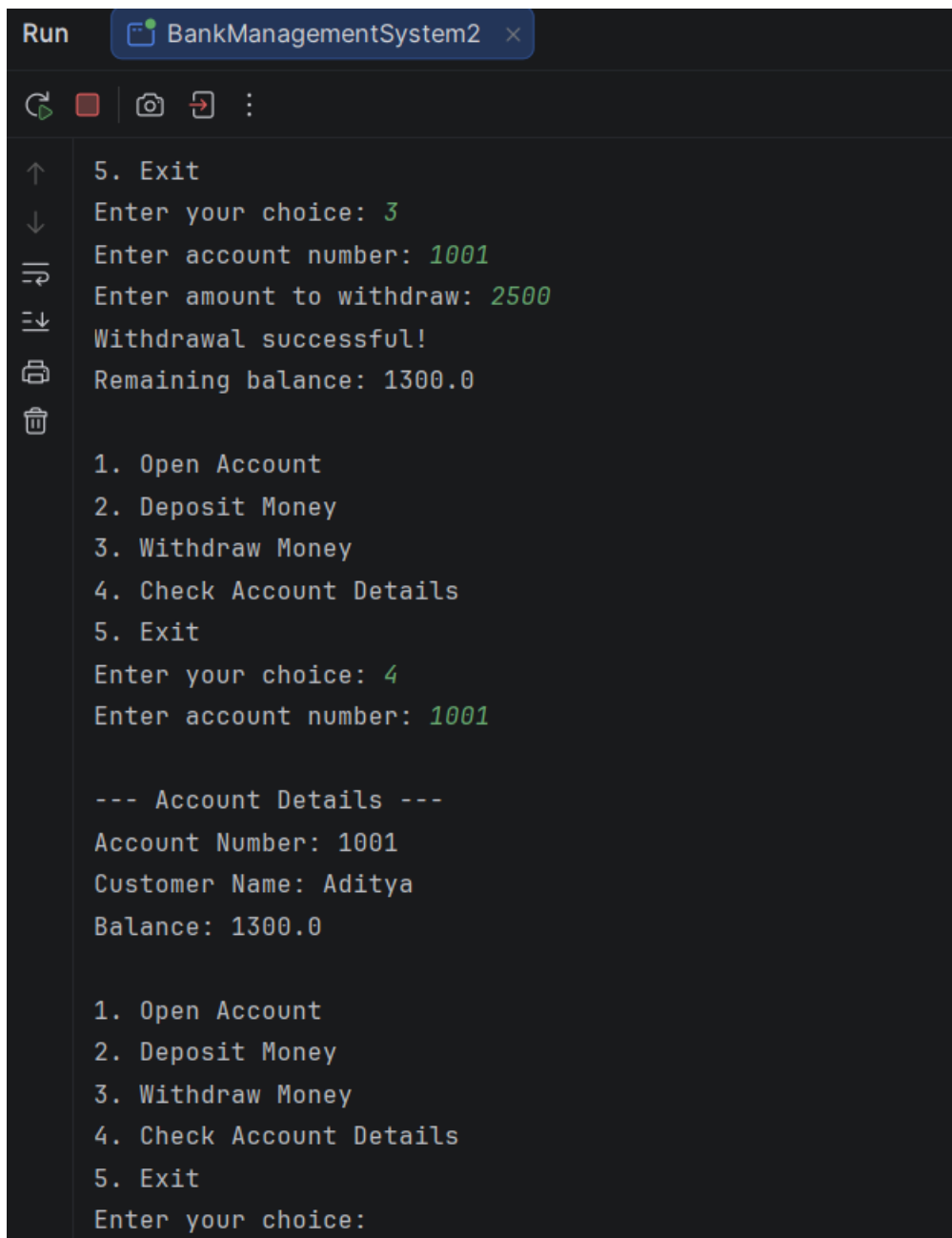
1. Open Account
2. Deposit Money
3. Withdraw Money
4. Check Account Details
5. Exit
```

Run

BankManagementSystem2 x



```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files  
==== Welcome to Bank App ====  
  
1. Open Account  
2. Deposit Money  
3. Withdraw Money  
4. Check Account Details  
5. Exit  
Enter your choice: 1  
Enter your name: Aditya  
Enter initial deposit: 2300  
Account created successfully!  
Your account number is: 1001  
  
1. Open Account  
2. Deposit Money  
3. Withdraw Money  
4. Check Account Details  
5. Exit  
Enter your choice:
```



```
Run BankManagementSystem2 x
5. Exit
Enter your choice: 3
Enter account number: 1001
Enter amount to withdraw: 2500
Withdrawal successful!
Remaining balance: 1300.0

1. Open Account
2. Deposit Money
3. Withdraw Money
4. Check Account Details
5. Exit
Enter your choice: 4
Enter account number: 1001

--- Account Details ---
Account Number: 1001
Customer Name: Aditya
Balance: 1300.0

1. Open Account
2. Deposit Money
3. Withdraw Money
4. Check Account Details
5. Exit
Enter your choice:
```

6. Limitations

Volatile Memory: The app uses a HashMap, so all your data disappears when you close the program.

Input Vulnerability: If you type a letter when it expects a number, the program will crash with an InputMismatchException.

7. Future Scope

File Handling: You could save account details to a text or CSV file so the data stays even after you close the app.

Database Integration: Connect it to a MySQL database with JDBC to manage data like a real application.

Exception Handling: Add try-catch blocks to handle bad user inputs without crashing.

Authentication: Make users enter a PIN or password to see their account details or withdraw money.

8. Conclusion

This project shows the fundamentals of Java programming. Building it taught me about variable scopes, loops, using the Scanner for input, and how to use HashMaps to link account numbers to account details. It's a good starting point for making more advanced programs that can save data.